

1. Information Representation

1.1. Data Representation

- The two fundamental characteristics of any number system are:
 - A base:** The number of different digits that a system can use to represent numbers
 - Place value:** The specific-value of a digit based on its position within a number
- Denary** - Base 10
- Binary Systems** - Base 2
 - Possible bits (binary digits): 0 and 1
 - All data and characters are represented in binary

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0

- For example, 65 in binary is *01000001*

Denary vs. Binary prefixes:

Denary Prefix	Factor Value	Binary Prefix	Factor Value
kilo- (k)	$\times 10^3$	kibi- (Ki)	$\times 2^{10}$
mega- (M)	$\times 10^6$	mebi- (Mi)	$\times 2^{20}$
giga- (G)	$\times 10^9$	gibi- (Gi)	$\times 2^{30}$
tera- (T)	$\times 10^{12}$	tebi- (Ti)	$\times 2^{40}$

Binary Coded Decimal (BCD)

- Binary representation where each positive denary digit is represented by a sequence of 4 bits (nibble).
- Only certain digits are converted to BCD because particular digits represent a digit greater than 9.
- For example, to represent 429 in BCD:
- Convert each individual digit of the number to its binary equivalents -
 - $4 = 0100$
 - $2 = 0010$
 - $9 = 1001$
- Concatenate the 3 nibbles (4-bit group) in order to produce BCD: *0100 0010 1001*
- Practical applications**
 - A string of digits on any electronic device displaying numbers (like in calculators)
 - Accurately measuring decimal fractions
 - Electronically coding denary numbers

Two's Complement

- We can represent a negative number in binary by making the most significant bit (MSB) a sign bit, which indicates whether the number is positive or negative.

Converting from negative denary to binary two's complement (example -42):

- Find the binary equivalent of the denary number (ignoring the negative sign) | $42 = 101010$
- Add extra 0 bits before the MSB, to format the binary number to 8 bits | 00101010
- Convert binary number to one's complement (flip the bits) | 11010101
- Convert binary number to two's complement (add 1) | $1010101 + 1 = 11010110$

Converting binary two's complement into denary (example 11010110):

- Flip all the bits | 00101001
- Add 1 | 00101010
- Convert binary to denary and put a negative sign) | -42

- Maximum positive number in 8 bits: 127
- Maximum negative number in 8 bits: -128

Hexadecimal Systems - Base 16:

- Possible digits: 0 to 9 and A to F, where A to F represent denary digits 10 to 15
- Practical applications:
 - Defining colours in HTML
 - Defining Media Access Control (MAC) addresses
 - Assembly languages and machine code
 - Debugging via memory dumps
- Example - A5 in Denary = $(16 \times 10) + (1 \times 5) = 165$
- Example - 65 in Hexadecimal = $65 \div 16 = 4$ | Remainder 1 | $\therefore = 41$

Character Sets:

- A character set generally includes upper & lower case letters, number digits, punctuation marks, and other characters.
- Character sets use different binary representations for each character via character encoding

Character Encoding Standards:

ASCII	Extended ASCII	Unicode
Only the English alphabet can be represented	ASCII's extension - Also includes most European languages' alphabets	Superset for ASCII & extended ASCII - recognized by various global languages
Each character encoding takes up 7 bits, hence 128 possible characters	ASCII extended to 8 bits, hence 256 possible characters	It has an excellent range of characters, using 2 or 4 bytes per character
Smaller storage space	2 or 4 times more storage space per character	

1.2. Multimedia - Graphics, Sound

• Bitmap Images

- *Definition:* Bitmap images are created by assigning a solid color to each pixel using bit patterns.
- *Encoding:*
 - Bit patterns are generated by considering each grid row as a series of binary color codes corresponding to each pixel's color.
 - These bit patterns are 'mapped' onto the main memory.
- *Pixels:* The smallest picture element whose color can be accurately represented by binary code.
- *File Header:* Bitmap images contain a file header with metadata, including image size, number of colors, etc.

• Image Resolution

- *Definition:* the number of pixels that make up an image. Example:- 4096x3192 pixels.
- *Effect:* Higher resolution results in sharper, more detailed images.

• Screen Resolution

- *Definition:* The number of pixels that can be viewed horizontally and vertically on a device's screen.
- *Calculation:* $\text{Number of pixels} = \text{width} \times \text{height}$ (e.g., 1680×1080 pixels).

• Colour depth

- *Definition:* The number of bits used to represent the colour of a single pixel.
- *Calculation:* An image with n bits has 2^n colours per pixel (e.g., a 16-color bitmap has 4 bits per pixel because $2^4 = 16$)
- *Effect:* Increasing color depth improves color quality but also increases file size.
- *File Size Calculation:*
 - File Size = Number of Pixels $\times$ Color Depth
 - Convert bits to bytes by dividing by eight if necessary.
- *Applications:* Scanned images and general computer usage, where small file sizes and easy manipulation are beneficial.

• Vector Graphics

- *Definition:* Made up of drawing objects.
- *Drawing Objects:* Mathematically defined constructs (e.g., rectangles, lines, circles).
- *Drawing List:* A set of commands defining the vector (e.g., points, lines, shapes, and their attributes that define the image.)
- *Properties:* Basic geometric data that determine the shape and appearance of each object.
- *Encoding:* Data is encoded using mathematical formulas to generate properties for drawing lines and curves to create the image.
- *Scalability:* Objects can be resized without losing quality, unlike bitmaps.
- *Applications:* Company logos and other graphics that require scaling.

• Sound

- *Analogue vs Digital:*
 - Analogue data consists of continuous electrical signals.
 - Digital data consists of discrete electrical signals.
- *Sound Signals:* Vibrations through a medium, inherently analogue due to infinite detail.
- *Conversion:*
 - Analogue signals are converted to digital signals by sampling.
 - The sound wave's amplitude is sampled at set time intervals.
 - These samples are encoded as a binary number sequence, providing a digital representation of the sound wave

- **Sampling Rate**
 - *Definition:* Number of samples taken per unit of time.
 - *Effect:* Increasing the sampling rate improves the accuracy of the digitized sound wave representation but increases file size.

- **Sampling Resolution**
 - *Definition:* Number of bits used to encode each sample.
 - *Effect:* Increasing sampling resolution improves the accuracy of digitized sound waves but increases file size.

- **Bit Rate**
 - *Definition:* Number of bits used to store 1 second of sound.
 - *Calculation:*
$$\text{Bit Rate} = \text{Sampling Rate} \times \text{Sampling Resolution}$$
 - *Length of Sound:* Measured in seconds.

1.3. Compression

- **Definition:** Compression is the process of reducing file size without significant loss in quality, resulting in:
 - Reduced time needed to search for data.
 - Faster transfer of compressed files, using less bandwidth than uncompressed files.

- **Lossless Compression**
 - *Definition:* A type of compression that allows original data to be perfectly reconstructed from a compressed file by utilizing some form of replacement.
 - *Examples:* Bitmap (.bmp), vector graphic (.svg), .png images, text file compression, database records.
 - *Run-Length Encoding (RLE):*
 - *Definition:* A form of lossless compression used for compressing text files and bitmap images.
 - *Mechanism:* Reduces file size by encoding sequences of adjacent, identical elements (characters in text files and pixels in bitmap images) into two values: run count and run value.
 - *Example:* RLE of a bitmap image.

- **Lossy Compression**

- *Definition:* A type of compression that irreversibly eliminates unnecessary data.
- *Effect:* File accuracy/quality is lower than with lossless compression, but file size is significantly reduced (often to about 10% of the lossless size).
- *Examples:* Sound files (.mp3), .jpeg images.
- *Mechanism in Sound Files:*
 - **Perceptual Coding:** Removes parts of the sound that are less audible or discernible to human hearing, as used in .mp3 compression.